

(a) Deterministic reference
(V1, single-threaded)



(b) V0 naive, 32 threads, RTX 4090
(race-corrupted; 21.5 dB /
SSIM 0.767 PSNR-Y vs. reference)



(c) V1 spatial reduction (CPU)
(bit-exact at every
thread count tested)



(d) V2 spatial reduction (GPU)
(bit-exact at every
grid config tested)

